The sharp corner threshold for five-colour cubes

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Abstract

Every collection of fifteen cubes, each displaying all five colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic exterior faces using those five colours. Fourteen need not suffice: a counterexample uses just three varieties, with multiplicities $4, 5, 5$. This determines the five-colour corner value listed as unknown by Berkove, Eickemeyer and Schulman (2018). The lower bound has an elementary proof. For the upper bound, anchored Hall deficiencies and a shared-corner argument give 357 finite cases. Containment, budget bounds and a deletion lemma give 279 smaller problems, each with a separately checked interval certificate. Lean proves the interval and packed checkers sound, verifies all 279 certificates and the fourteen-cube obstruction, and connects the finite problems to physical cube arrangements, yielding a complete formal proof of the sharp threshold. A reserve construction then shows that any n^3 five-colour cubes can all form a complete n-cube for every $n \geq 4$. The sharp corner thresholds with at most one, two, or at least three varieties are respectively $8, 13, 15$. These extensions are also formalised.

AI transparency. The research, code and manuscript were developed using GPT-6-Astra (OpenAI), through Codex, under the author's direction. This working manuscript has not received independent human peer review; Section 8 states the formal verification scope.

1 Introduction

MacMahon's coloured-cube puzzles [1] lead to a family of extremal questions: how many face-coloured cubes guarantee that eight can be chosen to make a larger cube with monochromatic faces? Each small cube here uses all colours of the specified palette. Colours remain fixed, proper rotations identify varieties, and reflections are not allowed as moves. Collections may contain repeated varieties. Only exterior faces are constrained; touching interior faces need not have equal colours.

Berkove, Eickemeyer and Schulman [2, Section 8, Table 7] tabulate the corresponding frame numbers for palettes of two to six colours and leave the five-colour column unknown. Their corner values for two, three, four and six colours are respectively $11, 23, 11, 24$. We determine the missing five-colour value; these thresholds need not increase with the palette size. For a $2 \times 2 \times 2$ cube the frame consists of its eight corners. Let N_5 denote the least integer such that every collection of N_5 five-colour cubes contains a solution using all five colours on the target.

Theorem 1. *The sharp five-colour corner threshold is* $N_5 = 15$.

Theorem 10 extends this result to frames of every side length and complete cubes of side length at least four, by an elementary reserve construction. Theorem 12 determines the corner threshold when the number of source varieties is bounded, and shows that the fourteen-cube obstruction has minimum support. The corner conclusion for larger collections follows by choosing any fifteen cubes. The theorem concerns a target that may depend on the collection; it does not assert the ability to construct every target.

The proof has two distinct parts. A short corner argument excludes every target for an explicit fourteen-cube collection. For the upper bound, Hall's theorem gives linear alternatives

for failure of a target. We cover hypothetical fifteen-cube counterexamples by 357 cases. Containment reduces these to 308, and initial budget bounds exclude another 29. A deletion lemma reduces the totals and budgets in the remaining 279 problems, which are excluded by finite interval certificates. The certificates, C++ programs, geometric audits, manuscript and Lean sources are included in the accompanying review archive.[1]

The companion paper [4] considers four-colour sources and targets using exactly three colours. It gives the sharp frame threshold $16n - 3$ for every side length $n \geq 2$, with a classification of the largest failing collections. Here both sources and targets use five colours, and the sharp corner result also gives the larger-cube conclusions above. The developments share the palette-independent geometry in `ColourCubes` [5]; neither threshold theorem is used to prove the other.

A limited literature search has found no later value for the five-colour entry. This is not a comprehensive priority claim. In particular, Rosenfeld's October 2026 article [6] has not been examined in full.

2 Geometry and Hall deficiencies

Represent a cube by its colours on the faces $(+x, -x, +y, -y, +z, -z)$ in a right-handed coordinate frame. A five-colour cube has one repeated colour. There are 75 varieties: for each repeated colour, three have its two faces opposite and twelve have them adjacent. In the opposite case, the other four colours form two opposite pairs, giving three possibilities. In the adjacent case, fix the two equal faces; the 24 orders of the other colours are identified in pairs by the rotation exchanging those equal faces. Thus there are 15 opposite-repeat varieties and 60 adjacent-repeat varieties. Each class is one orbit under colour permutations.

A corner type is the ordered triple of its incident face colours, modulo cyclic permutation. Order the outward face normals positively, equivalently read the triple counterclockwise from outside along the body diagonal. This reverses the clockwise convention in [2], consistently on every cube, and therefore preserves all matching and containment relations. Repeated occurrences of a corner type are retained as distinct target positions.

Lemma 2 (Corner–rotation correspondence)**.** *A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type at some corner.*

Proof. A proper rotation preserves the orientation of a corner frame, so it induces a cyclic permutation of its three outward faces. Conversely, any two positively oriented corner frames, with a prescribed cyclic alignment, are related by a determinant-positive signed permutation of the coordinate axes. This is a proper rotation of the cube. □

The palette-independent Lean version of this lemma is reused from the shared `ColourCubes` library [5].

Fix a target T. Form a bipartite graph whose left vertices are its eight corner positions and whose right vertices are the individual source copies. A copy is adjacent to a position when it fits there. A matching covering the eight positions is exactly a solution: each chosen copy can be rotated independently, since interior faces are unrestricted. By Hall's theorem [3], failure is equivalent to a nonempty set D of target positions having fewer than $|D|$ neighbouring copies.

Index the varieties by $i = 1, \ldots, 75$, and let x_i be their multiplicities. For a set S of varieties, write $x(S) = \sum_{i \in S} x_i$. Let $N_T(D)$ be the varieties fitting at least one position of D. Then failure of T is equivalent to the existence of a cut

$$x(N_T(D)) \leq |D| - 1, \qquad \varnothing \neq D \subseteq \{1, \ldots, 8\}. \tag{1}$$

There are 255 cuts per target, hence 19,125 cuts altogether.

[1] Intended repository: `https://github.com/jasondavies/five-colour-cubes`. Public release is pending; the review archive includes source checksums, the base Git revision and bundled pinned project dependencies.

3 An elementary fourteen-cube obstruction

Consider four copies of A, five of B, and five of C, where

$$A = (0, 0, 1, 2, 3, 4), \quad B = (0, 1, 2, 3, 2, 4), \quad C = (0, 3, 1, 2, 4, 1).$$

Their corner types are shown below. Triples denote cyclic classes; the last column counts positions whose type occurs in neither of the other varieties.

Variety	Corner types	Private positions
A	$013, 014, 023, 024, 031, 032, 041, 042$	5
B	$022, 023, 034, 042, 122, 124, 132, 143$	6
C	$011, 014, 021, 042, 113, 123, 134, 243$	6

Lemma 3. *The collection $4A + 5B + 5C$ has no five-colour solution.*

Proof. A solvable target must have every corner type in the displayed union. Its only two-colour types are $022, 122, 011, 113$. An adjacent-repeat target therefore repeats either colour 2, with types $022, 122$, or colour 1, with types $011, 113$. In either case these two corners determine the colours on the faces at the ends of the repeated-colour edge. The two orders of the remaining colours give a mirror pair: B and its mirror, or C and its mirror. The mirror of B requires 043, and the mirror of C requires 012; both types are absent. Only B, C remain.

For an opposite-repeat target, both orientations of each occurring unordered triple must be present. The unordered triples for which both cyclic orientations are available are

$$013, 014, 023, 024, 123, 134.$$

If its repeated colour is r, delete r from the four triples containing it. The remaining pairs must form a $K_{2,2}$ on the other four colours: choose one face from each of their two opposite pairs. Colours 2 and 4 occur in fewer than four listed triples. For $r = 1$ or $r = 3$, the four pairs have degree sequence $(3, 2, 2, 1)$, so do not form $K_{2,2}$. For $r = 0$, the pairs are $13, 14, 23, 24$, forcing opposite pairs $\{1, 2\}$ and $\{3, 4\}$. This is A.

Thus only A, B, C can be targets. Their private positions require respectively at least five, six and six copies of the corresponding variety, exceeding the available four, five and five copies. None is solvable. □

4 Reduction to finite problems

Eight copies of any variety solve that variety, so a solution-free collection has every multiplicity at most seven. Suppose such a collection has size 15.

Lemma 4 (Anchored Hall cover)**.** *Suppose a collection contains m copies of a variety A and does not solve A. There is a set K of its target positions such that*

$$|K| \le 7 - m, \qquad \#\{\textit{other copies fitting some position outside } K\} \le 7 - m - |K|. \tag{2}$$

Conversely, such a set certifies failure of target A.

Proof. Apply (1) to A and put $K = D^c$. Every copy of A is a neighbour of the nonempty set D. Subtract these m copies from the bound $|D| - 1 = 7 - |K|$. The converse is the same Hall inequality read backwards. □

First suppose the maximum multiplicity m is at least two. Choose an adjacent-repeat anchor attaining m if possible; otherwise choose an opposite-repeat anchor attaining m. In the latter case all adjacent-repeat multiplicities are at most $m-1$. Relabel the colours to send the anchor to a fixed representative of its orbit.

For each cover K in Lemma 4, put

$$S(K) = N_A(K^c) \setminus \{A\}, \qquad b(K) = 7 - m - |K|.$$

Write $S = S(K)$ and $b = b(K)$. The case has $x_A = m$, $x(S) \le b$, and the multiplicity bounds just described. Additional upper bounds use two-variety solutions. If a copies of A and $8-a$ copies of V build a target, with $a \le m$, then $x_V \le 7 - a$ in a solution-free collection. These tests are finite: the two varieties must cover all eight target positions, A must fit at least a, and V at least $8-a$. They are also sufficient: at most a positions are unavailable to V, and all of them are available to A. Assign these to A, extend the assignment to a positions that A fits, and assign the remaining $8-a$ positions to V.

Colour permutations and simultaneous reflection of the entire collection preserve solvability. First identify equal pairs $(S(K), b(K))$, even when they arise from inequivalent position subsets K; then retain one representative per orbit of these pairs under the anchor stabiliser. The reported counts are constraint-orbit counts, not position-subset counts. Transporting a cover also transports its constrained varieties, budget and multiplicities, so no counterexample is lost. Global reflection is used only to transport a whole hypothetical counterexample, not as an allowed move on an individual cube. The repeated-variety cases have the following counts, totalling 244.

Anchor / maximum multiplicity	2	3	4	5	6	7
Adjacent-repeat	72	55	31	14	4	1
Opposite-repeat	26	21	11	6	2	1
Total	98	76	42	20	6	2

For distinct collections a different choice of anchor reduces the count.

Lemma 5 (Shared two-colour corner)**.** *Among eleven adjacent-repeat cubes, two share a two-colour corner type. If a distinct collection contains such a pair and one member is chosen as anchor, the anchor's two two-colour positions cannot be Hall-deficient.*

Proof. Each adjacent-repeat cube has exactly two such types, *aab* and *aac*, with $b \ne c$. There are only $5 \cdot 4 = 20$ types *aab*. Eleven cubes contribute 22 incidences, so two cubes share a type. For the two characteristic positions of the chosen anchor, the anchor itself and the other cube are two distinct neighbours. Their neighbourhood therefore has size at least two. □

If an opposite-repeat variety is present, choose it as anchor; this gives 31 constraint orbits with every multiplicity at most one. Otherwise use Lemma 5 to choose an adjacent-repeat anchor. Omit the cover complementary to its two characteristic positions and set all opposite-repeat capacities to zero. Its stabiliser preserves these positions, so the omission is compatible with the orbit reduction. It removes one of the 83 adjacent-repeat constraint orbits, leaving 82 further cases.

Proposition 6. *Every solution-free fifteen-cube collection is carried by colour relabelling and, if needed, global reflection into one of the 357 exported root cases.*

Proof. The preceding arguments cover repeated and distinct collections. The finite catalogue, two-variety bounds and coverage of all anchor-cover orbits are checked from physical rotations by the accompanying audit. It regenerates rotations from quarter turns, separately from the

signed-axis permutations used by the case generator, and checks every exported bound and every cover orbit. The resulting counts are

$$244 + 31 + 82 = 357.$$

The cases need not be disjoint; completeness of the cover is what is required. □

Each root specifies an anchor A of multiplicity m, coordinate bounds ℓ, u, a set S of non-anchor varieties and a budget b. The bounds fix $\ell_A = u_A = m$, with $\ell_i = 0$ for $i \neq A$. A vector $x \in \mathbb{Z}^{75}_{\geq 0}$ is *admissible* for this root if

$$\begin{gathered} \ell \leq x \leq u, \qquad \sum_{i=1}^{75} x_i = 15, \qquad x(S) \leq b, \\ \text{for every target } T, \text{ there is } \varnothing \neq D \subseteq \{1, \ldots, 8\} \text{ with } x(N_T(D)) \leq |D| - 1. \end{gathered} \tag{3}$$

Thus admissibility includes failure of every target, in addition to the anchor constraints.

The cover can be reduced without further search. If $S' \subseteq S$ and $b' \geq b$, then $x(S) \leq b$ implies $x(S') \leq b'$ for nonnegative x. Together with coordinatewise inclusion of the root boxes, this identifies 49 cases contained in other cases, leaving 308. Write u for the root upper bounds. In 29 retained cases, $b + u(S^c) < 15$ already contradicts the prescribed total, so only 279 cases need further exclusion proofs. The exact map and retained case numbers are in `data/q5-root-reduction.json`; Lean checks all containments and arithmetic exclusions. The archive retains all 357 original cases.

Lemma 7 (Hall-budget projection)**.** *Suppose a root has anchor multiplicity m, budget $x(S) \leq b$, and total 15, with the anchor outside S. For $0 \leq s \leq b$, put $G = 15 - b + s$ and suppose $G \geq m$. Every admissible vector x has a vector $y \leq x$ with the same anchor multiplicity, total G, budget $y(S) \leq s$, and a Hall deficiency for every target. It obeys the original upper bounds, tightened to $\min(u_i, s)$ on S.*

Proof. First retain only $\min(x(S), s)$ copies in S. This removes at most $b - s$ copies and leaves the anchor untouched, so the remaining total is at least G. Delete further non-anchor copies until the total is G; this is possible because $G \geq m$. Every coordinate decreases, so all upper bounds and Hall deficiency inequalities are preserved. Nonnegativity and $y(S) \leq s$ give $y_i \leq s$ for $i \in S$. □

For our anchored covers, $b = 7 - m - |K|$, so

$$G = m + 8 + |K| + s.$$

Thus the reduced collection consists of the m fixed anchor copies and $8 + |K| + s$ other copies; in particular, $G \geq m$ is automatic. Apply the lemma after selecting and normalising the original root, retaining its bounds. No new anchor selection is required for the smaller collection. Of the 279 remaining roots, 277 admit an exclusion certificate with $s = 0$. The other two, original cases 68 and 69, use $s = 1$ and $G = 12$. Their $s = 0$, $G = 11$ problems actually have solution-free collections, supplied and checked by physical matching in the projected manifest. The projected totals range from 9 to 15. All 101 roots whose original budget was zero retain their original certificates.

5 Interval certificates

For a projected problem, admissibility is defined by (3) with the tightened upper bounds, total G and budget s. We certify that no admissible vector exists.

Write a box as $\ell \le x \le u$. For a cut $x(S) \le r$, two necessary conditions for feasibility in the box are

$$\ell(S) \le r, \qquad G \le r + u(S^c). \tag{4}$$

The first follows from the lower bounds; the second follows from $G = x(S) + x(S^c)$. When they hold, put $s = r - \ell(S)$ and $g = r + u(S^c) - G$. Every satisfying vector obeys

$$x_i \le \min(u_i, \ell_i + s) \quad (i \in S), \qquad x_i \ge \max(\ell_i, u_i - g) \quad (i \notin S). \tag{5}$$

For example, the upper bound follows by giving all other coordinates of S their lower bounds; the lower bound follows by giving all other coordinates of S^c their upper bounds.

For a target, retain its cuts satisfying (4). If none remain, no vector in the box can fail that target. Otherwise let I and U be the intersection and union of their neighbourhoods, and let s_*, g_* be the largest corresponding s, g. Then all vectors in the box that fail this target remain in the refined box

$$u_i' = \begin{cases} \min(u_i, \ell_i + s_*) & i \in I, \\ u_i & i \notin I, \end{cases} \qquad \ell_i' = \begin{cases} \max(\ell_i, u_i - g_*) & i \notin U, \\ \ell_i & i \in U. \end{cases} \tag{6}$$

Indeed, a failing vector satisfies one of the retained cuts, and its bounds in (5) imply (6). The root budget is treated as a family containing a single cut. The total $\sum_i x_i = G$ gives

$$\ell_i' = \max\left(\ell_i, G - \sum_{j \ne i} u_j\right), \qquad u_i' = \min\left(u_i, G - \sum_{j \ne i} \ell_j\right).$$

A total-bound leaf closes exactly when $\sum_i \ell_i > G$ or $\sum_i u_i < G$. A budget leaf closes when $\ell(S) > b$ or $b + u(S^c) < G$; a target leaf closes when none of its 255 cuts satisfies (4). These are the checker's 77 possible closing reasons. Refinements must tighten at least one bound and leave nonempty coordinate intervals; a split must satisfy $\ell_i \le v < u_i$.

A certificate is a finite tree of refinements and integer splits $x_i \le v$ or $x_i \ge v + 1$. Each leaf is excluded by total bounds or by a cut family with no surviving cut. Induction on this tree proves that its root contains no admissible vector. The separate C++ checker reads these instructions and verifies the deductions; it performs no search and does not use the search program's matching implementation.

Proposition 8 (Certificate replay)**.** *The separate checker accepts an exclusion certificate for each of the 279 projected problems.*

The projected archive contains 499,829 instructions in 313,094 compressed bytes, reducing the instruction count by 95.69% from the original archive. The repository documents both archives and the earlier search computations. Every archived header is checked against its exported root case. Eighty negative controls are rejected, including all 77 possible direct closing claims against a separate $G = 11$ fixture, not a $G = 15$ root problem. The witness satisfies its total, fixed anchor count, coordinate bounds and zero root budget, and fails all 75 targets by physical matching. The fixture is `data/q5-distinct-known-control.txt`; its regeneration and witness checks are in `scripts/check_five_colour_distinct.py`.

Proof of Theorem 1. Lemma 3 gives $N_5 \ge 15$. A solution-free collection of fifteen cubes would satisfy an exported root case by Proposition 6, and would supply a Hall deficiency for every target. Containment and the initial budget tests leave one of 279 roots. Lemma 7 supplies an admissible vector in its projected problem, which Proposition 8 and the sound interval deductions exclude. Hence $N_5 \le 15$. □

6 From corner solutions to larger cubes

An n-frame consists of the corners and edges of an $n\times n\times n$ cube, and uses $8+12(n-2) = 12n-16$ pieces. Let $F_5(n)$ be the least pool size guaranteeing an n-frame with monochromatic exterior faces displaying all five colours. The frame may use a subset of the pool.

Lemma 9 (Reserve extension)**.** *For every integer $q \geq 1$, any collection of five-colour cubes has a subcollection of at most $25q-4$ cubes such that every five-colour corner solution formed from the remaining cubes extends, using the reserved cubes, to a $(q+2)$-frame.*

Proof. An edge type is an unordered pair of face colours, with repetition allowed. A five-colour target with repeated colour r has at most one monochromatic edge, necessarily rr. Each bichromatic edge type occurs at most twice; if it occurs twice, it contains r, since every other colour occupies just one face. A source whose repeated colour is b has an ab edge for every $a \neq b$: its a-face cannot be opposite both b-faces.

For each colour a, reserve up to q cubes having an aa edge, taking all such cubes if fewer exist. Write R_a for this reserve, and call a *full* if $|R_a| = q$. These reserves are disjoint, since a source has only one repeated colour. If k colours are full, their total size is at most $5q-5+k$.

Next process the ten bichromatic pairs ab in any order. From the still-unreserved cubes having an ab edge, reserve q if both a, b are full, and $2q$ otherwise; if fewer remain, reserve all of them. Denote this reserve by E_{ab}. All reserves are disjoint, and together have size at most

$$5q-5+k+20q-q\binom{k}{2} \leq 25q-4,$$

since $q \geq 1$ and $k-\binom{k}{2} \leq 1$ for $0 \leq k \leq 5$.

Take a corner solution using only unreserved cubes. Every edge type of its target occurs on a chosen corner cube. Hence each reserve for an occurring edge type reached its requested size: otherwise its reservation step would have removed every remaining cube with that edge type. In particular, if the target has an rr edge, R_r is full; use it for that edge's q non-corner positions.

Fill each bichromatic edge from its dedicated reserve. A type occurring once needs q cubes, and a type occurring twice needs $2q$. The only shortfall is a doubled type rb for which both colours are full, so $|E_{rb}| = q$. Use E_{rb} for one edge and R_b for the other. Every cube in R_b has an rb edge by the first paragraph. Different doubled pairs use different R_b, and none uses R_r. Thus the allocations are disjoint. Any matching pair of adjacent faces can be rotated to a prescribed edge, so these assignments extend the corner solution to the frame. □

Theorem 10. *For every integer $n \geq 3$,*

$$12n-16 \leq F_5(n) \leq 25n-39.$$

For every integer $n \geq 4$, any collection of n^3 five-colour cubes can all be assembled into an $n \times n \times n$ cube with monochromatic exterior faces displaying all five colours.

Proof. The frame's piece count gives the lower bound. Put $q = n-2$. A pool of $25q+11$ leaves at least fifteen cubes outside the reserve of Lemma 9. Theorem 1 supplies corners from these, and the reserve supplies the edges. This proves the upper bound.

For $n \geq 4$, $n^3 \geq 25n-39$: at $n=4$ this is $64 \geq 61$, and for $n \geq 5$ it follows from $n^2 \geq 25$. Construct a frame from the n^3 cubes. Each remaining face-interior position prescribes just one colour, which every unused cube has, and interior positions prescribe none. Fill these positions with all remaining cubes. □

In particular, $F_5(3) \leq 36$ and $F_5(4) \leq 61$, while any 64 cubes form a complete $4\times4\times4$ cube without spares. The least pool size for a complete n-cube is therefore exactly n^3 for $n \geq 4$. This argument does not decide whether every collection of 27 cubes forms a complete 3-cube.

7 Collections with few varieties

Let $M(k)$ be the least number of cubes guaranteeing a corner solution when the collection contains at most k varieties.

Lemma 11. *Two five-colour varieties with disjoint corner sets can build an opposite-repeat target from four copies of each.*

Proof. This is a small finite geometric check. Of the $\binom{75}{2}$ unordered pairs, 330 have disjoint corner sets. Up to colour permutation and exchange of the sources, they have the following five representatives. The first source is 010234 throughout; tuples use the face order of Section 2. Each listed target has four corners supplied by each source.

Second source	Target
011243	001234
012324	012234
012334	012234
012434	012234
030412	001234

The accompanying geometric checker regenerates all varieties and verifies the orbit coverage and each witness. Lean checks a suitable target directly for every pair, without using the orbit reduction. □

Theorem 12. *For every integer $k \geq 1$,*

$$M(k) = \begin{cases} 8 & k = 1, \\ 13 & k = 2, \\ 15 & k \geq 3. \end{cases}$$

Consequently, three is the minimum number of varieties in a fourteen-cube collection without a solution.

Proof. Eight identical copies build their own variety, and fewer than eight pieces cannot form a corner solution. For two varieties, consider thirteen copies. If either variety occurs at least eight times, it builds itself. Otherwise the multiplicities are seven and six. If the varieties share a corner type, use a minority copy at that corner of the majority target and seven majority copies elsewhere. If their corner sets are disjoint, apply Lemma 11.

For the lower bound take six copies each of

$$P = (0, 1, 0, 2, 3, 4), \qquad Q = (0, 1, 1, 3, 2, 4).$$

Their corner sets share just 041. Inspection of the 75 varieties shows that only P, Q have all corner types in their union. Each has seven private positions, so neither can be built with six copies. The geometric checker verifies these assertions; Lean certifies failure using deficient corner sets for all 75 targets. Thus $M(2) = 13$. For $k \geq 3$, Theorem 1 supplies the upper bound, and the three-variety collection in Lemma 3 supplies the lower bound. □

Remark (A family of fourteen-cube obstructions). Keep A, C from Section 3, and put

$$B_1 = (0, 1, 2, 3, 2, 4), \quad B_2 = (0, 1, 2, 3, 3, 4), \quad B_3 = (0, 1, 2, 4, 3, 4).$$

All 21 collections

$$4A + 5C + x_1B_1 + x_2B_2 + x_3B_3, \qquad x_1 + x_2 + x_3 = 5, \quad x_i \in \mathbb{Z}_{\geq 0},$$

are unsolvable. The B_i share the six trichromatic corner types $023, 034, 042, 124, 132, 143$. The combined corner union allows only A, C, B_1, B_2, B_3 as targets, as a check of the catalogue confirms. Target A needs five A-copies, C needs six C-copies, and any B_i needs six copies from the B-group. The respective supplies are four, five and five. Lean verifies these shortages simultaneously for every distribution. The original example is $x_1 = 5, x_2 = x_3 = 0$.

8 Formalisation and reproducibility

The package uses Lean 4 and mathlib [7, 8], both at version `v4.35.0`. Lean proves soundness of the reference interval checker, of a faster arithmetic implementation, and of composition of bounded proof chunks. The faster implementation encodes a multiplicity vector as

$$P(x) = \sum_{i=0}^{74} x_i B^i, \qquad B = 1024.$$

For a neighbourhood S, the weight $W(S) = \sum_{i \in S} B^{74-i}$ satisfies

$$x(S) = \left\lfloor \frac{P(x)W(S)}{B^{74}} \right\rfloor \bmod B.$$

Lean proves this identity under $0 \le x_i \le 7$, including the absence of carries; the coefficient bound is $75 \cdot 7 < 1024$. It also proves decoding, coordinate updates and equivalence of the packed feasibility test to the integer inequalities. All 19,125 supplied weights are checked against their neighbourhood masks by the Lean kernel.

Chunk composition requires exact agreement of both boundary vectors and a proof excluding each referenced box; the exporter is untrusted. A limit of 128 instructions gives 4,786 chunks for the projected archive. All 279 projected certificates have passed kernel replay, together with their final assembly into the physical threshold and sharpness theorems. The exact case mapping and certificate hashes are in `data/q5-projected-root-manifest.json`. The 93 declarations listed in `Audit.lean` and the final replay and extension theorems have audited dependencies confined to `propext`, `Classical.choice` and `Quot.sound`; no `native_decide` is used.

Lean proves completeness and uniqueness of the 75 cube varieties and checks all 45,000 compatibility bits using the shared library's palette-independent corner–rotation theorem. The physical lower-bound theorem, `FiveColour.Physical.lower_bound` in `FiveColour/LowerBound.lean`, uses 75 deficient corner sets: 72 singletons and three sets of sizes 5, 6 and 6. In the formal proof, this finite check replaces the elementary target classification in Lemma 3; catalogue completeness and rotation transport extend it to every physical target.

Lean proves the physical Hall reduction, symmetry transport, anchor normalisation, two-variety bounds, anchor selection and the shared-corner argument. Finite checks certify the root capacities and cover representatives. Containment and initial budget tests reduce the 357 roots to 279. Lemma 7 is formalised by `FiveColour.Physical.root_projection` in `FiveColour/RootProjection.lean`; all concrete projection hypotheses and coverage of the retained roots are kernel-checked.

The projected archive's integer semantics are established in `FiveColour/ProjectedBridge.lean`. The identifications include all coordinate bounds, prescribed totals, budgets and Hall-cut families. This checks the entire admissibility predicate, not just the packed boundary vectors. The generated module `FiveColour/ProjectedChunks/Threshold.lean` assembles all 279 exclusions into the unconditional theorems `FiveColour.Packed.ProjectedChunks.threshold15` and `FiveColour.Packed.ProjectedChunks.sharpThreshold15`.

The reserve argument is formalised in `FrameReserve.lean`, using generic lemmas from `Reserve.lean`; both files are in `FiveColour/`. A finite check of 75 targets and 32 full-colour

patterns supplies the edge allocation rule; the reservation and allocation proofs apply to arbitrary q. Completion of a frame is proved in `FiveColour/FullCube.lean`, adapting the general construction from [4] without using its threshold theorem. `FiveColour/SmallSupport.lean` proves the two-variety bound and its sharpness; `FiveColour/MixedObstructions.lean` proves the whole obstruction family by common Hall deficiencies. These additions need no new interval certificates.

The unconditional conclusions in `Solution.lean`, in namespace `FiveColour.Puzzle`, are `frame_bound`, `full_cube`, `complete_cube_threshold` and `support_threshold`. Their axiom checks, including the 64-cube instance, are listed in `ExtensionsAudit.lean`.

The original and intermediate interfaces are documented in `FiveColour/README.md`. Reproduce the full proof and its axiom audit with:

```
make check WORKERS=2
make projected-replay-data WORKERS=8
make projected-replay WORKERS=8
make check-extensions
```

The review archive supplies certificates, code and reproduction commands. A Mathlib-only comparator challenge states the puzzle geometrically. Model equivalence is proved by `FiveColour.Puzzle.hasSolution_iff` in `FiveColour/ComparatorBridge.lean`. The solution module applies the completed threshold theorem and builds in Lean. A successful full external comparator check has not yet been confirmed. Mathematical questions left open here include classifying all fourteen-cube obstructions and determining the sharp five-colour frame numbers for $n \geq 3$. In particular, do twenty cubes always suffice for a 3-frame, and can every collection of 27 cubes form a complete 3-cube?

AI disclosure and review status

GPT-6-Astra (OpenAI), used through Codex, developed the arguments, search and verification programs, Lean formalisation and manuscript under the human author's direction. The author initiated and directed the research. Author review and submission approval of this working manuscript are pending. The inherited geometry has its own provenance in the pinned dependency. Automated checks and AI review are not independent human peer review.

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